

CHAPTER 18

Spacecraft operations

18.1. Space story

I was at the Charles Stark Draper Laboratory (CSDL) during the launch of the Space Shuttle with the first communications-satellite payload. CSDL designed the Space Shuttle control systems. The communications satellite was mounted on a spin table that would spin it up before separation. It was already in orbit when my boss came to me and asked me to see if the bias momentum would have any effect on the Orbiter control. I ran simulations and the control system worked just fine. In the meantime, my boss did a simple calculation in his head and computed the possible torque from the angular rates of the Orbiter and determined that the $\omega \times h$ torque was insignificant. This shows why having a firm knowledge of physics is so important!

18.2. Introduction

Most people are familiar with images of NASA and JPL mission-control centers. Engineers sit at consoles where they can observe data sent from orbiting satellites. Operators can send commands to the spacecraft. The spacecraft or rovers operated at JPL are robotic. Those operated from Baikonur, NASA/JSC, or NASA/KSFC involve people onboard the spacecraft.

18.3. Preparing for a mission

Preparations for a mission are very similar to preparations for live opera, ballet, or play. A satellite mission is no different from a live theater. Once the performers are on stage they must deal with what happens in real-time. There are no “do-overs.” Spacecraft missions are much the same. Preparation for a mission is modeled after theater, as shown in Fig. 18.1.

Fig. 18.2 shows the typical organization for a launch team. In some organizations, this team is permanent and their only job is to operate spacecraft. In others, the subsystem team member joins the launch-support organization only for the launch. Though nominal lines of communication are shown, it is not unusual for subsystem engineers to engage other subsystem engineers when issues arise. The attitude control and mission-planning groups often have the same manager because their functions are tightly interrelated. The Shift Supervisor is usually one of the more experienced subsystem engineers. Working as a shift supervisor is a good way to help develop system engineers.

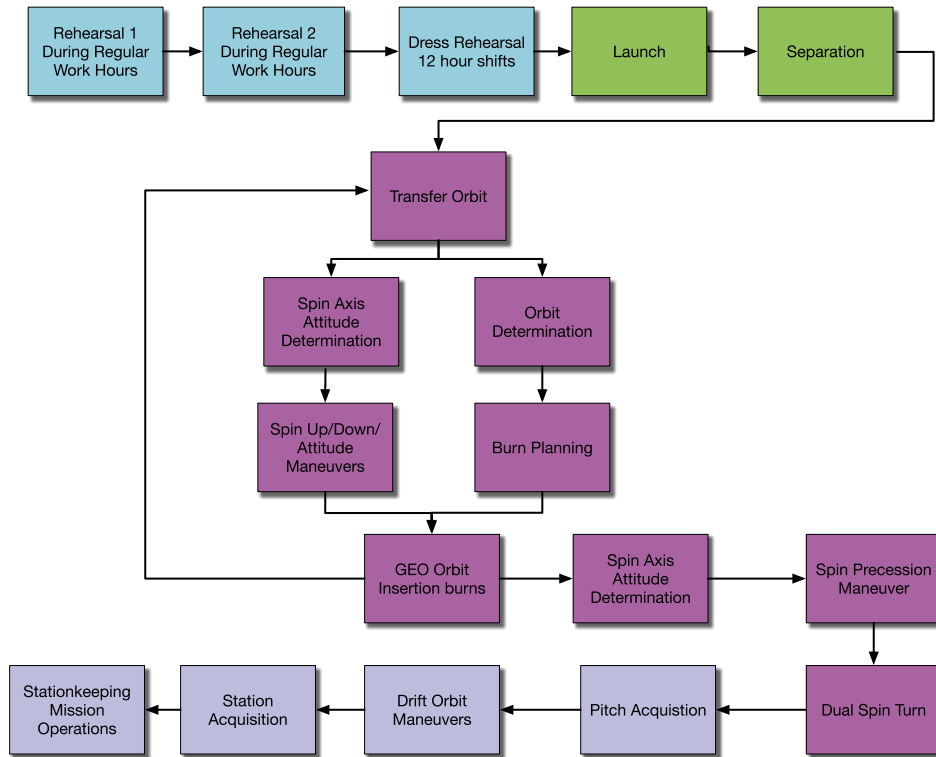


Figure 18.1 Rehearsal schedule for a satellite launch.

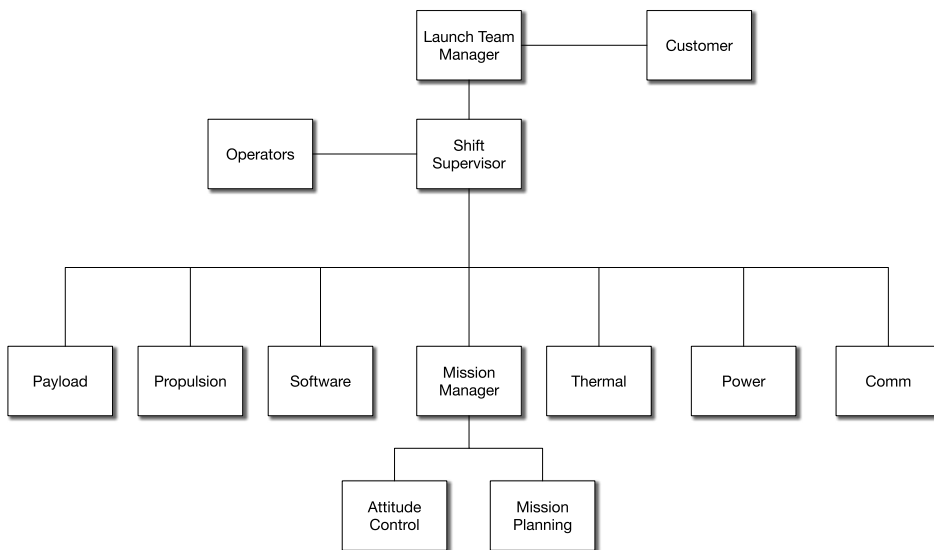


Figure 18.2 Satellite-launch team.

18.4. Elements of flight operations

Elements of mission operations are shown in Fig. 18.3. The satellite is controlled by the flight software and the flight software is commanded via the communication links from the control center. Both the flight and ground software require databases of parameters that include parameters for the flight software, telemetry-frame definitions, command databases, and ground-software parameters. The ground software is used by the flight-operations team to perform analyses, create command sequences, and compute the trajectory and the required orbit changes. The line between the databases and

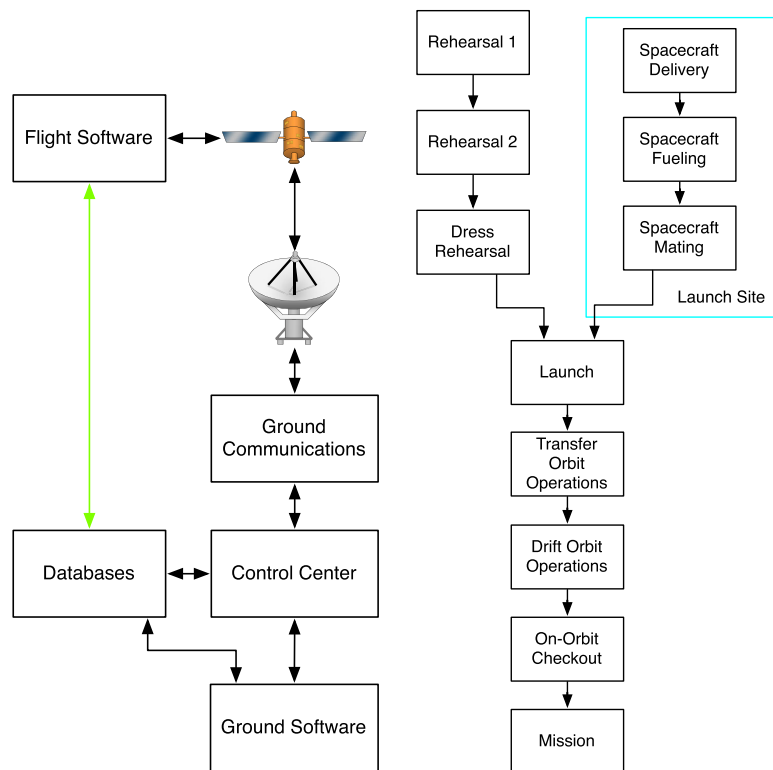


Figure 18.3 Mission-operations elements and timeline.

flight software is virtual. There is no direct link but the databases hold all changeable parameters in the flight software. In some cases, an actual database is a resident on the flight processor but sometimes the parameters are distributed throughout the software itself.

18.5. Mission-operations timeline

The mission-operations timeline is shown in Fig. 18.3. At the end of manufacturing and integration and test, the spacecraft is shipped to the launch site. In parallel, the mission-operations team prepares for the mission operations through a series of rehearsals.

18.6. Mission-operations entities

Many organizations are involved in a satellite launch. An example of the entities and their connectivity is shown in Fig. 18.4. The mission-control center is the center of the launch operations and is supported by several other organizations. The method of connectivity was once the commercial phone system but now is the internet. Management above the level of the operations team may maintain a physical presence. Management only gets involved in the event of serious anomalies. The customer will always have representatives at the control center during the launch but may rely on telecommunications later in the mission. In commercial operations, the customer often has their own control center. NORAD can provide initial orbital elements since they track all satellites.

The operations team communicates with the satellite via ground antennas and the tracking stations. Each tracking station can monitor telemetry and send commands should the phone system go down during critical operations.

Connections to the launch site include the satellite-launch team and the rocket-launch team. The rocket-launch team is in complete control until satellite separation and rarely needs feedback from the satellite-control team.

All entities connect through the communications system. However, not all entities can perform all functions or communicate directly with all other entities. For example, only the control center can send commands to the spacecraft or obtain telemetry. The customer usually cannot talk with the spacecraft until handover occurs.

The data rates of the various links vary greatly. The links to the spacecraft generally have low data rates, on the order of 2 kbps. For deep-space spacecraft, the data rates may be as low as bits/second. Links between entities on the ground can reach the maximum speed possible with ground communications.

18.7. Mission-operations preparation

Once the satellite is complete and ready to ship to the launch site the mission-operations teams begin preparations. The following items need to be completed before the first rehearsal.

1. Telemetry database;
2. Command database;
3. Spacecraft alignments;
4. Spacecraft-mass properties;

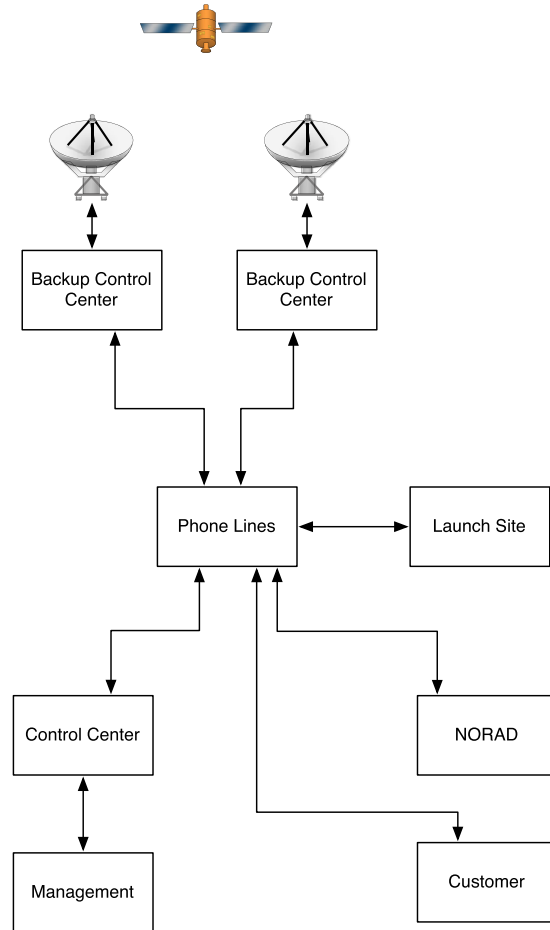


Figure 18.4 Entities involved in mission operations.

5. Spacecraft-component locations;
6. Calibrations of sensors and actuators;
7. Mission-maneuver plan;
8. Mission timeline;
9. Onorbit checkout plan;
10. Procedures handbook.

18.8. Mission-operations organization

The mission-operations organization is shown in Fig. 18.5. Spacecraft usually require 24-hour coverage during critical mission phases that include launch, transfer orbit, and

early mission-orbit operations. This is not always possible for spacecraft that do not have multiple ground stations or intersatellite links, such as to TDRS. The launch team is organized into shifts with a mission director supervising the shift supervisors. Shifts can range from 8 to 12 hours. Under the shift supervisor are the subsystem leads and the spacecraft operator. The spacecraft operator sends commands to the spacecraft. While the software is not a subsystem there are always ground- and flight-software engineers to handle any software problems that may come up during operations. The software engineers also handle computer and networking issues both on the ground and in space. The mission-planning team performs orbit determination and determines when to fire

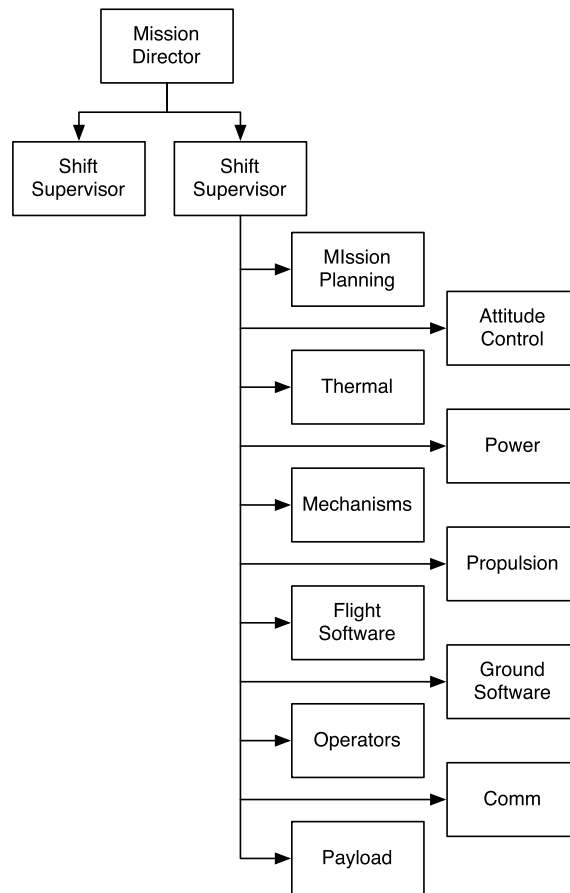


Figure 18.5 Mission-operations organization.

the thrusters to progress the spacecraft towards its final orbit. They deliver burn plans, which include thrusters to be used, burn duration, and burn start time, to the attitude control engineers who plan the maneuver. Modern spacecraft sometimes have GPS

systems or other autonomous navigation systems so they may not need ground-based orbit determination.

The attitude control engineers are responsible for the pointing of the spacecraft during all mission operations. In addition, they are responsible for the momentum state of the spacecraft and schedule momentum unloading if needed. The attitude control engineers track the fuel consumption of the attitude control thrusters, even when they are used for orbit-change maneuvers or station keeping.

The thermal engineers track spacecraft temperatures. They command heaters on or off as needed to maintain the temperatures of all components within the desired bounds.

The power engineers monitor the power state of the satellite. They command charging rates for the batteries and may command offsets of the solar wings if too much power is being absorbed.

The mechanisms engineers are primarily responsible for deployments of antennas, solar wings, etc. They usually are the engineers to command the detonation of pyro-electric bolt-cutters, etc.

The propulsion engineers are responsible for the propellant-distribution systems, that is to say, the plumbing, onboard the spacecraft. They command the firing of pyro valves that connect the fuel tanks to the thrusters. The attitude control engineers usually are responsible for the engines themselves.

The flight- and ground-software engineers correct problems inflight and ground software. The former may be fixed by uploading patches to the flight software.

The operators issue all commands to the spacecraft. The commands may be stored in lists written by subsystem engineers but the operators send each command and verify that they have been correctly received by the spacecraft.

18.9. Mission-control center

A modern mission-operations center is organized as shown in Fig. 18.6. Redundancy is not shown but the server and network would all be redundant. Generally, at least two possible ground stations are available.

18.10. Mission-operations example

A typical station-keeping maneuver is described below. It is assumed that at each step the next step is approved.

1. Mission analysts use range and range data to produce an up-to-date set of osculating orbital elements. If the spacecraft has GPS, this may be done onboard.
2. Mission analysis plans a velocity change to correct the orbit.
3. The plan is given to the shift supervisor who confers with the mission director.

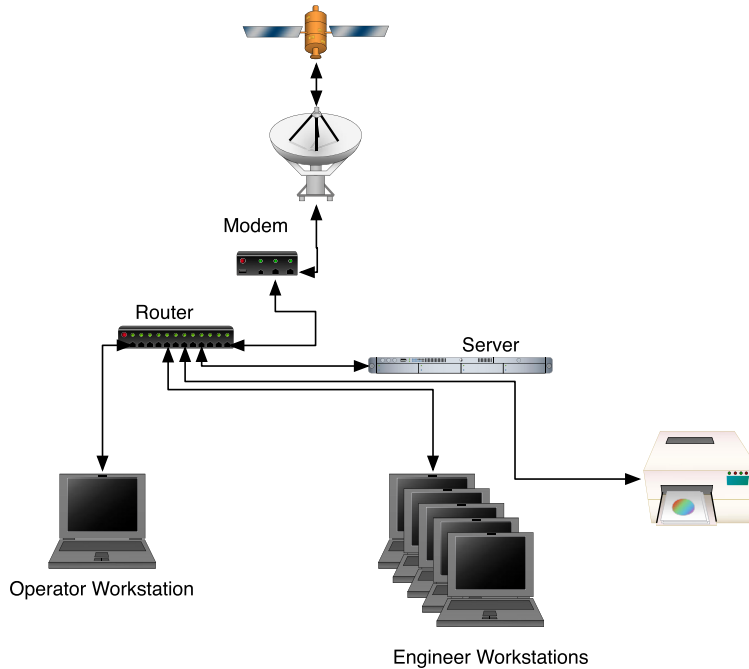


Figure 18.6 Mission operations.

4. The approved plan is given to attitude control. The attitude control analysts prepare the command lists. The spacecraft will transition into station-keeping mode, thrusters and gyros need to be enabled and gyro and thruster heaters enabled to warm up the devices.
5. Thermal and power engineers review the timing of the plan to ensure that no temperature or power limits will be exceeded.
6. Thermal and power engineers submit command lists to the operator to adjust battery rate of charge and heater status. They monitor telemetry to verify the successful completion of the commands.
7. The spacecraft operator instructs the ground station to increase transmission power.
8. The attitude control engineers submit a command list to enable thrusters and gyros. They monitor telemetry to verify the successful completion of the commands.
9. The attitude control engineers submit a command list to load station-keeping parameters. They monitor telemetry to verify the successful completion of the commands.
10. The station-keeping abort command list is preloaded into the command consoles at the command center and supporting ground stations.
11. The attitude control engineers submit a command list to start station keeping.

12. All subsystem engineers monitor their subsystems to make sure all telemetry stays within desired limits.
13. Station keeping terminates. Gyros and thrusters are disabled.
14. Attitude control engineers monitor transition into normal mode.
15. Transmission power is reduced.
16. The operator begins periodic range and rate collections.
17. Mission analysts compute the new orbit using assumed burn parameters. The actual burn parameters will be estimated as part of orbit determination.

For many organizations, commands need to be approved by one or more layers of management. This may reduce errors but will also slow down reaction time to unforeseen events. In some cases, the decision to send a command may require days of meetings. With the advent of sophisticated integrated simulations, simulations may be conducted to verify an operation sequence before operating.